Smart Retail Analytics & Product Recommendation System

Project Overview

The Smart Retail Analytics & Product Recommendation System is a Data Science project designed to analyze retail transaction data, identify purchasing patterns, and generate product recommendations using Association Rule Mining techniques.

The system integrates data preprocessing, exploratory data analysis, recommendation generation, FastAPI backend services, and an interactive Streamlit dashboard.

Problem Statement

Retail businesses generate thousands of transactions daily. Understanding customer purchasing behavior can help increase sales, improve product placement, and provide personalized recommendations.

This project aims to:

- Discover frequently purchased product combinations.
- Generate product recommendations.
- Analyze revenue and sales trends.
- Provide business insights through an interactive dashboard.

Dataset

Dataset: Online Retail Dataset

Features:

- InvoiceNo
- StockCode
- Description
- Quantity
- InvoiceDate
- UnitPrice
- CustomerID
- Country

Project Workflow

Data Collection

The Online Retail dataset was used as the primary data source.

Data Cleaning

Steps performed:

- Removed missing values
- Removed duplicate records
- Removed cancelled orders
- Removed negative quantity transactions

Feature Engineering

Created:

Revenue = Quantity × UnitPrice

Additional time-based features were generated for sales analysis.

Exploratory Data Analysis

Performed:

- Top Selling Products Analysis
- Revenue Analysis
- Country-wise Sales Analysis
- Monthly Sales Trends

Association Rule Mining

Apriori Algorithm

Used to identify frequent itemsets and association rules.

Metrics:

- Support
- Confidence
- Lift

FP-Growth Algorithm

Used as an optimized alternative to Apriori for large datasets.

Recommendation Engine

A recommendation system was developed using association rules generated from the Apriori model.

Example:

Input: LUNCH BAG RED RETROSPOT

Output: LUNCH BAG SUKI DESIGN

FastAPI Backend

Endpoints:

- /
- /recommend
- /health
- /business-insights

Streamlit Dashboard

Features:

- KPI Dashboard
- Product Analytics
- Market Basket Analysis
- Product Recommendations

Business Insights

Examples of discovered associations:

- Customers purchasing LUNCH BAG RED RETROSPOT frequently purchased LUNCH BAG SUKI DESIGN.
- Customers purchasing PINK REGENCY TEACUP AND SAUCER frequently purchased GREEN REGENCY TEACUP AND SAUCER.

These insights can be used for:

- Product bundling
- Shelf optimization
- Cross-selling strategies

Technologies Used

- Python
- Pandas
- NumPy
- Plotly
- Streamlit
- FastAPI
- Apriori
- FP-Growth
- Mlxtend

Conclusion

The Smart Retail Analytics & Product Recommendation System successfully demonstrates how Association Rule Mining and Retail Analytics can be combined to generate business insights and product recommendations. The project showcases end-to-end Data Science workflow from data preprocessing to deployment-ready analytics.